Student Placement Data Analysis Using Exploratory Data Analysis (EDA)

1. Introduction

Student placement plays a significant role in evaluating the academic quality and career readiness of students. Educational institutions continuously strive to improve placement opportunities by understanding the factors that influence employability. Academic performance alone is no longer sufficient, as employers also consider communication skills, practical experience, project work, and extracurricular activities.

This project aims to perform **Exploratory Data Analysis (EDA)** on a student placement dataset to identify the major factors affecting placement outcomes. By analyzing various student attributes, meaningful insights can be generated to help institutions improve training programs and provide better career guidance. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used in Google Colab for data preprocessing and visualization.

2. Data Collection

The dataset used for this project was obtained from Kaggle, a popular platform for open-source datasets and machine learning projects. The dataset contains student-related information that influences placement outcomes.

Dataset Attributes

- CGPA
- Previous Semester Result
- Academic Performance
- Communication Skills
- Internship Experience
- Projects Completed
- Extracurricular Activities
- Placement Status

The dataset was imported into **Google Colab** for preprocessing and analysis using Python.

3. Data Preprocessing

Before performing data analysis, several preprocessing techniques were applied to improve the quality of the dataset.

3.1 Data Loading

The dataset was loaded into Google Colab using the Pandas library.

3.2 Dataset Inspection

The following checks were performed:

- Displayed the first few records using head()
- Checked dataset dimensions using shape
- Examined column names
- Verified data types using info()
- Generated summary statistics using describe()

3.3 Missing Value Analysis

The dataset was checked for missing values using:

- isnull().sum()

No significant missing values were found.

3.4 Duplicate Record Check

Duplicate records were identified using:

- duplicated().sum()

Duplicate entries were removed to ensure data consistency.

3.5 Data Validation

The CGPA column was examined carefully because valid CGPA values should not exceed **10**.

Records containing invalid CGPA values greater than 10 were removed from the dataset to improve data accuracy.

3.6 Data Encoding

Categorical values such as **Yes** and **No** were converted into numerical values (**1** and **0**) to simplify analysis and visualization.

3.7 Outlier Analysis

A box plot was used to examine the CGPA distribution. Since no significant outliers were observed after removing invalid values, no further outlier treatment was required.

4. Exploratory Data Analysis (EDA)

Several visualization techniques were used to understand the relationship between different student attributes and placement status.

4.1 Placement Distribution

Visualization: Count Plot

Purpose:

- To determine the number of placed and non-placed students.

Observation:

- The distribution of placed and non-placed students was clearly identified.

4.2 CGPA Distribution

Visualization: Histogram

Purpose:

- To observe the distribution of student CGPA.

Observation:

- Most students had a CGPA between **7 and 8**, indicating good academic performance.

4.3 Communication Skills vs Placement

Visualization: Bar Chart

Purpose:

- To compare average communication skills between placed and non-placed students.

Observation:

- Students with better communication skills generally showed better placement outcomes.

4.4 Internship Experience vs Placement

Visualization: Count Plot

Purpose:

- To study the influence of internship experience on placement.

Observation:

- Students with internship experience had better placement opportunities.

4.5 Projects Completed vs Placement

Visualization: Count Plot

Purpose:

- To analyze the relationship between completed projects and placement.

Observation:

- Students completing more academic projects showed improved placement performance.

4.6 Academic Performance Distribution

Visualization: Histogram

Purpose:

- To examine the academic performance of students.

Observation:

- Most students demonstrated moderate to good academic performance.

4.7 Correlation Heatmap

Visualization: Heatmap

Purpose:

- To identify relationships between numerical features.

Observation:

- Positive correlations were observed between placement status and attributes such as CGPA, communication skills, and academic performance.

4.8 Placement Percentage

Visualization: Pie Chart

Purpose:

- To display the overall percentage of placed and non-placed students.

Observation:

- The chart provided a clear representation of placement outcomes.

5. Outcome of the Analysis

The exploratory data analysis revealed several important insights:

- Students with higher CGPA generally achieved better placement outcomes.
- Good communication skills positively influenced placement success.

- Internship experience significantly improved employability.
- Students completing more academic projects demonstrated better placement performance.
- Academic performance contributed positively to placement opportunities.
- Placement success depends on a combination of academic knowledge, practical experience, and soft skills rather than a single factor.

6. Recommendations

Based on the analysis, the following recommendations are suggested:

- Students with lower CGPA should receive additional academic support and mentoring to improve their performance.
- Students actively involved in extracurricular activities should be encouraged to explore career opportunities that match their interests and talents.
- Students who possess strong communication skills but relatively lower academic performance should receive career guidance for communication-oriented roles such as customer support, sales, business development, and client relations.
- Students with good academic records and multiple completed projects but who remain unplaced should receive industry-oriented project training, resume-building sessions, mock interviews, and placement preparation programs.
- Educational institutions should increase internship opportunities, technical workshops, and skill-development programs to improve student employability.
- Personalized career counseling should be provided to help students identify their strengths and select suitable career paths.

7. Conclusion

This project successfully analyzed student placement data using Exploratory Data Analysis techniques in Python. Through data preprocessing, visualization, and interpretation, several factors influencing placement outcomes were identified. The analysis showed that academic performance, communication skills, internships, and project experience collectively contribute to placement success. The findings and recommendations from this study can help educational institutions design effective training programs and improve overall student employability.